Safety Helmet Detection Using YOLOv8

AIRI Team PITB AI Internship — Computer Vision Task 1

1. Project Objective

The objective is to build an object-detection model that identifies people or objects in the two configured classes: With Helmet and Without Helmet. The project covers dataset preparation, training, evaluation, and visual inspection of predictions.

2. Dataset and Split

A 250-image subset was selected from a larger publicly available Roboflow helmet-detection dataset. The original dataset is credited as the source; the project does not claim to have collected the original images itself. The subset preserves separate train, validation, and test splits and uses YOLO-format label files.

Split	Images	Share
Train	175	70%
Validation	50	20%
Test	25	10%
Total	250	100%

3. Model and Training Setup

• Model: YOLOv8n with pretrained weights

• Training duration: 30 epochs

• Image size: 640 pixels

• Batch size: 16

• Compute environment: Google Colab with NVIDIA T4 GPU

• Best checkpoint: runs/helmet_250images/weights/best.pt

4. Test Evaluation

The following metrics were obtained on the held-out 25-image test split. Because the test set is small, these scores should be interpreted as an initial experiment rather than a guarantee of real-world performance.

Metric	Test result
Precision	83.2%
Recall	82.5%
mAP@50	84.6%
mAP@50–95	54.0%

Class	Precision	Recall	mAP50	mAP50–95
With Helmet	74.1%	91.3%	84.0%	59.5%
Without Helmet	92.3%	73.7%	85.2%	48.6%

5. Training and Evaluation Visuals

The following figures are included when found in the saved training and evaluation folders.

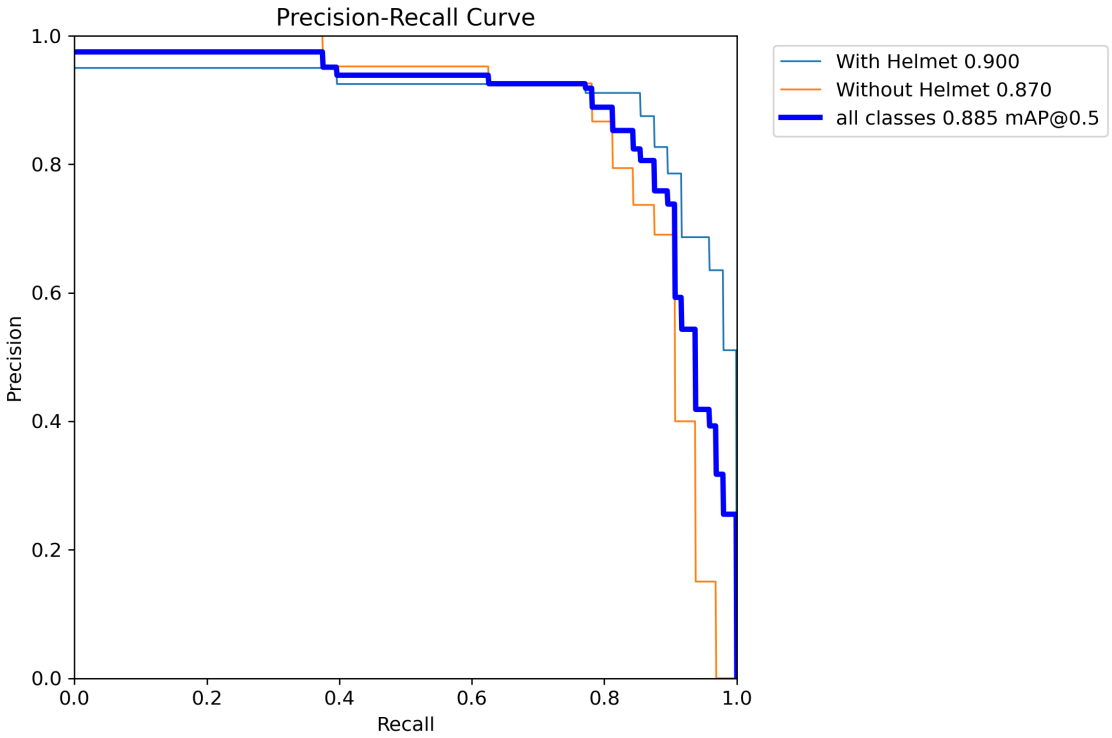
val_batch0_pred.jpg



val_batch0_labels.jpg



BoxPR_curve.png



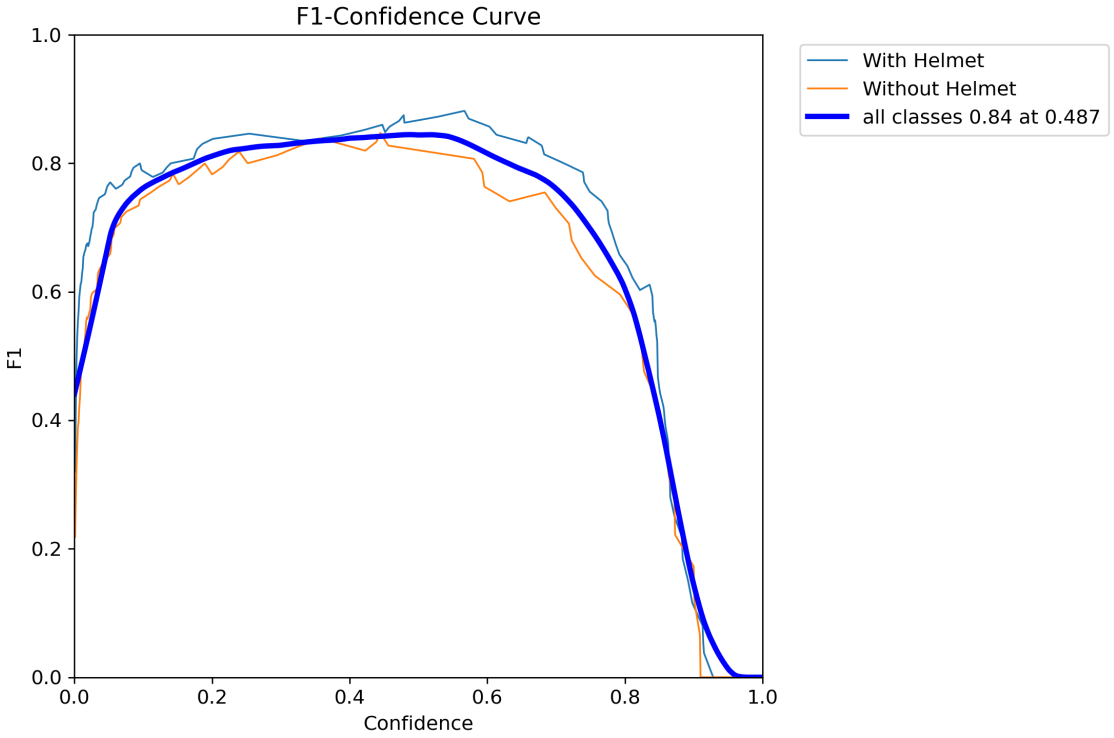
val_batch1_labels.jpg



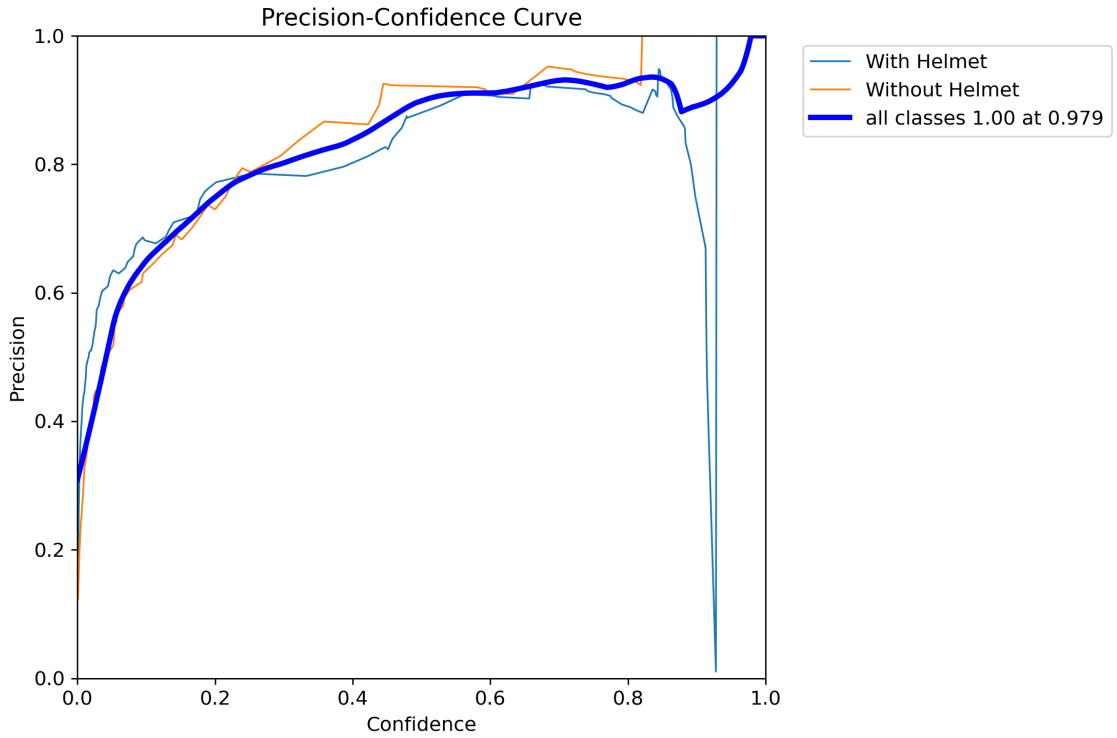
val_batch1_pred.jpg



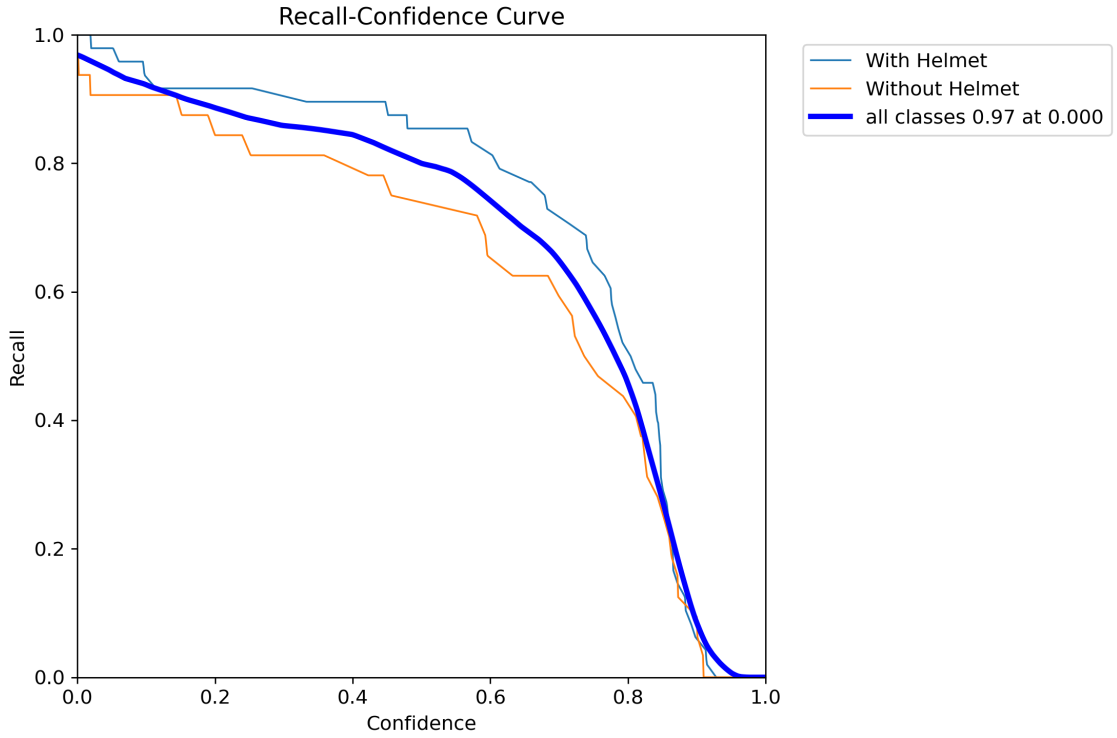
BoxF1_curve.png



BoxP_curve.png



BoxR_curve.png



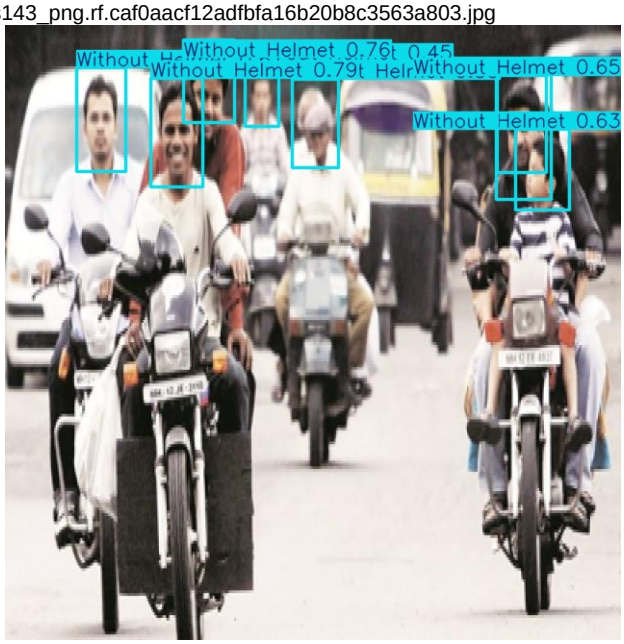
6. Prediction Examples

Found 15 saved prediction images. Up to the first 15 are included below as visual examples. Prediction images show model outputs; they do not by themselves prove every detection is correct.

Prediction example 1: BikesHelmets102_png.rf.b771e83db4052e5e48900f1d9f4ee68e.jpg



Prediction example 2: BikesHelmets143_png.rf.caf0aacf12adfbfa16b20b8c3563a803.jpg



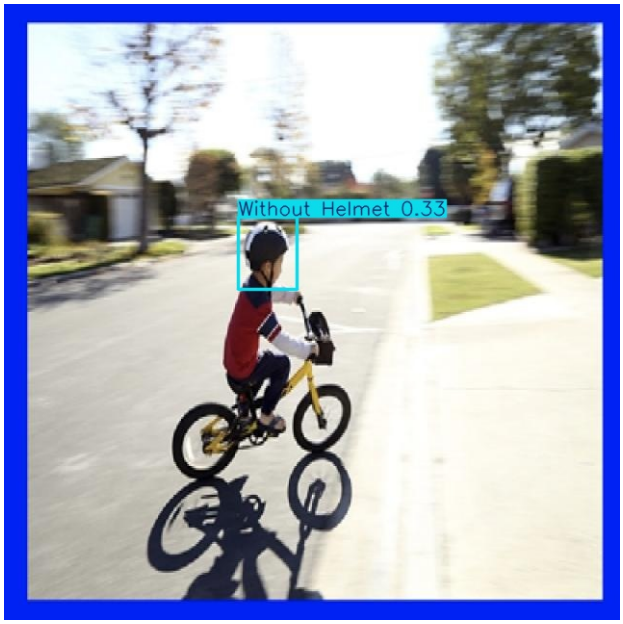
Prediction example 3: BikesHelmets158_png.rf.71cdb3536b840ccc2f0e44d1e5ce36e3.jpg



Prediction example 4: BikesHelmets183_png.rf.1f9ebed2be11011e589ec58d1def89c1.jpg



Prediction example 5: BikesHelmets214_png.rf.1d87985c9f70bdff315954906fd27b10.jpg



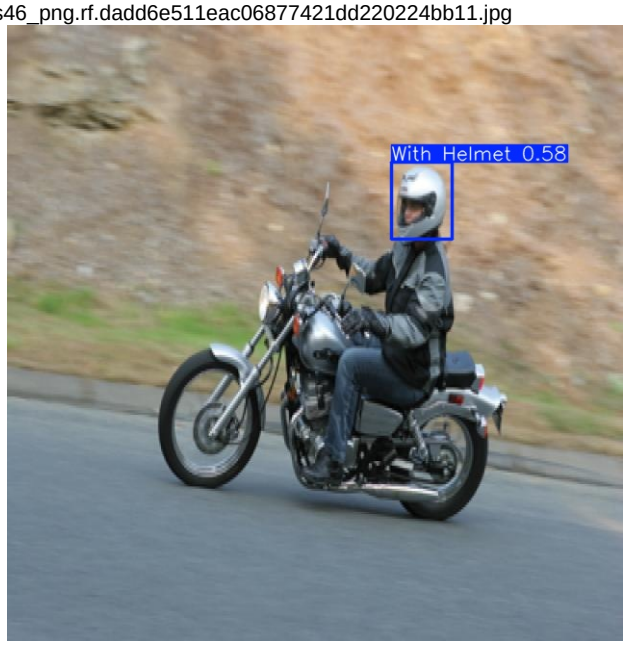
Prediction example 6: BikesHelmets367_png.rf.28ba774989591d521958acedb1b16f60.jpg



Prediction example 7: BikesHelmets425_png.rf.fa3787994b86131edecb1298002e5d05.jpg



Prediction example 8: BikesHelmets46_png.rf.dadd6e511eac06877421dd220224bb11.jpg



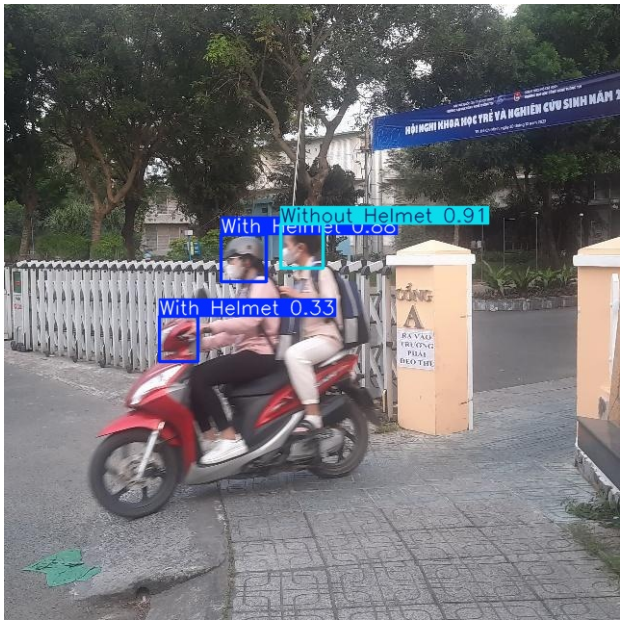
Prediction example 9: BikesHelmets690_png.rf.82cdd08be3388e9ac5b4d508f4036c1d.jpg



Prediction example 10: BikesHelmets746_png.rf.2c1f249429743284635e7469d5917206.jpg



Prediction example 11: test16_jpg.rf.d10e44bdf0099562590be3254b0d7f26.jpg



Prediction example 12: test28_jpg.rf.5721290a682f2de37d19ac2d21eb68de.jpg



Prediction example 13: test37_jpg.rf.04195d3807dec2cd84c84ace5989848c.jpg



Prediction example 14: test47_jpg.rf.135f45bbf540aba11476f4ab13c19532.jpg



Prediction example 15: train147_jpg.rf.d149be3d2280f0d73ba7ca93b5f66e15.jpg



7. Error Analysis and Limitations

Ten test examples were visually reviewed, and no obvious errors were noticed in those examples. This is limited visual inspection, not proof that the model has no errors. The test set contains only 25 images. The project should separately document at least 10 genuinely weak or incorrect predictions, such as missed objects, false detections, or low-confidence cases, after inspecting the actual images and labels. Do not label a prediction as wrong without verifying it against the ground-truth annotation.

Potential cases for further review

- Small or distant helmets that may be difficult to localise.
- Partially occluded people or helmets.
- Low-light images and unusual camera angles.
- Low-confidence detections or missed objects.
- Possible confusion between the two classes.

8. Conclusion and Future Work

The YOLOv8n model achieved 84.6% mAP@50 on the selected test split. Next steps include verifying annotations on at least 100 images, documenting 10 actual weak/error cases, expanding the dataset, and testing on diverse real-world footage. This prototype should not be used as the sole basis for workplace-safety decisions.

9. LinkedIn-Ready Summary

Built a Safety Helmet Detection prototype using YOLOv8n. Prepared a 250-image dataset subset, trained the model for 30 epochs on Google Colab, and evaluated it on a held-out test split, achieving 84.6% mAP@50. The project includes training artefacts, evaluation results, and prediction visualisations. Further annotation review and error analysis are planned.